Zexin Jian

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Education Background

National University of Defense Technology (NUDT) **Changsha, China**
B.E. Student, School of Computer Science *Sept.2021 – present*
Advised by Dr. Zengkun

Chinese Academy of Sciences (CAS) **Beijing, China**
Visiting Student, Institute of Computing Technology (ICT) *Jun.2024 – present*
co-Advised by Dr. Huimin Cui and Dr. Chenxi Wang

Research & Publications

- **[NeurIPS'24 CCF-A]** *Perception of Knowledge Boundary for Large Language Models through Semi-open-ended Question Answering*, Zhihua Wen, Zhiliang Tian, **Zexin Jian**, Zhen Huang, Pei Ke, Yifu Gao, Minlie Huang, Dongsheng Li.
- **[ACL'23 CCF-A]** *GRACE: gradient-guided controllable retrieval for augmenting attribute-based text generation*, Zhihua Wen, Zhiliang Tian, Zhen Huang, Yuxin Yang, **Zexin Jian**, Changjian Wang, Dongsheng Li.

Projects

NUDT SysY-Compiler *Feb.2024 – Aug.2024*

- Developed a fully functional compiler system for SysY2022 (a C Subset) using C++ as the primary programming language.
- Over 50,000 lines of code, including 30,000 lines of handwritten code and 20,000 lines auto-generated using Python based on custom templates.

NUDT RustOS *Sept.2024 – Dec.2024*

- The starting point of this project is to explore the impact of different kernel architectures on kernel security and performance and try to balance the two.
- Starting from the macro kernel, absorbing and learning from the design patterns of the micro kernel to improve security while ensuring certain performance.

Awards and Honors

- 2022年全国大学生数学竞赛解放军赛区一等奖;
- 2023年全国大学生数学建模竞赛湖南省赛区三等奖;
- 2023年全军军事建模竞赛二等奖;
- 2023年全国大学生数学竞赛解放军赛区三等奖;
- 2024年第十五届蓝桥杯湖南省省二等奖;
- 2024年程序设计天梯赛, 个人全国铜牌, 团队湖南省铜牌;
- 2024年编译系统实现赛全国二等奖;
- 2024年红山开源操作系统竞赛一等奖;
- 国防科技大学学习标兵/国防科技大学计算机学院院三好学生

Skills

- Programming Languages: C/C++, Python, Rust.
- Research Interests: Managed Runtime System, Machine Learning System, LLM.
- IT Skills: Linux, PyTorch, CMake.